Q6 NO CALCULATOR ALLOWED Q6

Answer QUESTION 6 PARTS A and B on this page.

PART A

$$\lim_{x\to\infty}\left|\frac{\frac{(x-4)^{n+2}}{(n+2)3^{n+1}}}{\frac{(x-4)^{n+1}}{(n+1)3^n}}\right| \quad \lim_{n\to\infty}\left|\frac{(x-4)^{n+2}}{(n+2)3^{n+1}}\cdot\frac{(n+1)(3^n)}{(x-4)^{n+1}}\right|$$

$$\lim_{n\to\infty}|x-4| \qquad -1 < x-4 < 1$$

$$+4 \qquad +4 \quad +4$$

$$3 < x < 5 \qquad (3,5)$$

$x=3$

$$\lim_{n\to\infty}\frac{(-1)^{n+1}}{(n+1)(3^n)}$$ ~~Converges~~

PART B

$$f'(x) = \frac{(x-4)^2}{12} + \frac{(x-4)^3}{81} + \frac{(x-4)^4}{16\cdot 3^3} + \ldots + \frac{(x-4)^{n+1}}{(2(n+1))3^n}$$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

0070656

Q5525/14

Answer QUESTION 6 PARTS C and D on this page.

PART C

$$\lim_{x \to \infty} \left| \frac{\frac{(x-4)^{n+2}}{(2n+4)3^{n+1}}}{\frac{(x-4)^{n+1}}{(2n+2)3^{n}}} \right| \quad \lim_{x \to \infty} \left| \frac{(x-4)^{n+2}}{(2n+4)3^{n+1}} \cdot \frac{(2n+2)3^{n}}{(x-4)^{n+1}} \right| \quad \lim_{x \to \infty} \left|\right.$$

PART D

No, since $x = 8$ is not in the interval of convergence, which is $(3,5)$, the Taylor series cannot converge at that value.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.